

DIARIZATIONLM: SPEAKER DIARIZATION POST-PROCESSING WITH LARGE LANGUAGE MODELS

Quan Wang* Yiling Huang* Guanlong Zhao*
 Evan Clark Wei Xia Hank Liao
 Google LLC, USA *Equal contribution
 ✉ quanw@google.com

ABSTRACT

In this paper, we introduce DiarizationLM, a framework to leverage large language models (LLM) to post-process the outputs from a speaker diarization system. Various goals can be achieved with the proposed framework, such as improving the readability of the diarized transcript, or reducing the word diarization error rate (WDER). In this framework, the outputs of the automatic speech recognition (ASR) and speaker diarization systems are represented as a compact textual format, which is included in the prompt to an optionally finetuned LLM. The outputs of the LLM can be used as the refined diarization results with the desired enhancement. As a post-processing step, this framework can be easily applied to any off-the-shelf ASR and speaker diarization systems without retraining existing components. Our experiments show that a finetuned PaLM 2-S model can reduce the WDER by rel. 55.5% on the Fisher telephone conversation dataset, and rel. 44.9% on the Callhome English dataset.

Code: <https://github.com/google/speaker-id/tree/master/DiarizationLM>

Model: <https://huggingface.co/google/DiarizationLM-8b-Fisher-v2>

Demo: <https://huggingface.co/spaces/diarizers-community/DiarizationLM-GGUF>

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1 INTRODUCTION

Speaker diarization is the task of partitioning speech into homogeneous segments according to speaker identities, answering the question “who spoken when” [1, 2]. Typical speaker diarization systems can be roughly categorized into two groups: modularized systems and end-to-end systems. A modularized speaker diarization system usually consists of multiple separately trained components including voice activity detection (VAD) [3, 4, 5, 6], speaker turn detection [7, 8], speaker encoder [9, 10, 11], and a clustering algorithm, which can be either unsupervised [12, 13, 14, 15, 16, 17] or supervised [18, 19]. End-to-end systems, on the other hand, directly optimize the entire system on diarization errors by introducing a permutation invariant loss function [20, 21, 22, 23].

In many real world applications such as meeting summarization, call center analysis, mobile recorder apps [24], and video captioning, knowing “who spoke when” is not sufficient. Speaker labels are more interpretable and meaningful when they are associated with speech transcripts. Various solutions have been proposed to directly address the problem of “who spoke what”, including jointly training speech recognition and speaker diarization [25], speaker-attributed automatic speech recognition (SA-ASR) [26, 27, 28, 29], target speaker automatic speech recognition (TS-ASR) [30, 31, 32, 33] and word-level end-to-end neural speaker diarization [34].

In practice, however, most production speech systems still consist of separately trained ASR models and speaker diarization models, with various considerations including:

1. *Modularized development and deployment:* ASR and speaker diarization systems are usually trained on different datasets, and potentially using different modeling framework, by different research teams.

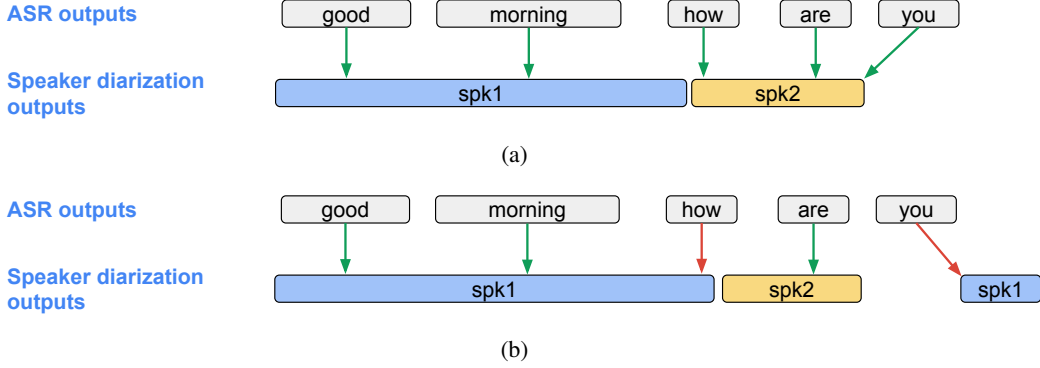


Figure 1: The orchestration module associates each word from the ASR transcript with a speaker label from the speaker diarization outputs. (a) In this example, all words are associated with the correct speaker labels (green arrows). The words “good”, “morning”, and “are” and “you” are associated with the only speaker label that overlap with them. The word “how” overlaps with both spk1 and spk2, but has bigger overlaps with spk2, thus is associated with spk2. The word “you” does not overlap with any speaker, but is closest to spk2, thus is associated with spk2. (b) In this example, two words are associated with wrong speaker labels (red arrows) due to inconsistent timing information from the two systems. The word “how” is mistakenly associated with spk1, since spk1 has more overlap with this word than spk2. The word “you” is mistakenly associated with spk1, since spk1 is closer to this word than spk2.

2. *Potential quality regression on ASR*: ASR has many more use cases than speaker diarization. Joint modeling of ASR and speaker diarization usually has worse Word Error Rates (WER) than ASR-only models, thus is not acceptable in many applications.
3. *Flexibility*: Combining separately trained ASR models and speaker diarization models is a very flexible solution. As long as the ASR model provides word timing information, it can be combined with almost any speaker diarization model, either unsupervised or supervised, either modularized or end-to-end trained.

We refer to the combination of ASR transcripts and speaker diarization results as an *orchestration module* (in some other work [35], this process is called “reconciliation”). In this module, each word from the ASR transcript is associated with a speaker label. A typical orchestration algorithm works as follows: (1) If the word segment overlaps with at least one speaker segment, then this word is associated with the speaker that has the biggest temporal overlap with this word; (2) otherwise if this word segment does not overlap with any speaker segment, then it is associated with the speaker that has the smallest temporal distance to this word based on the segment boundaries. This orchestration algorithm is illustrated in Fig. 1a.

However, since ASR and speaker diarization are separately trained with usually different training datasets and modeling approaches, the timing information from these two systems can be inconsistent, resulting in word diarization errors, as demonstrated with the example in Fig. 1b. Specifically, modern ASR models are usually trained end-to-end without using the ground truth timing information, and the word timing is inferred from the probability lattice of the decoder, which could be inaccurate.

In many cases, such errors can usually be fixed by leveraging semantic information from the ASR transcripts. Take Fig. 1 as an example, simply by looking at the textual transcript “good morning how are you”, if we know it consists of two speakers, we can easily tell which word comes from which speaker confidently without using any acoustic speaker diarization system. In practice, diarization errors can be much more complicated than the simple example in Fig. 1. To handle such cases, we propose DiarizationLM, a framework to post-process the orchestrated ASR and speaker diarization outputs with a large language model (LLM). While the experiments performed in this paper mainly focus on reducing word diarization errors using LLM, we also show examples how LLM can be used for other purposes such as autofilling speaker names, autofilling speaker roles, or even completely replacing the orchestration module in Appendix B.